

TED UNIVERSITY

CMPE 491 / Senior Project

RAG-Based Turkish Question Answering and Source Citing Document Assistant

Project Proposal

Fall 2026

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Project Name	RAG-Based Turkish Question Answering and Source-Citing Document Assistant
Project URL	https://esimayc.github.io/tr-rag-website/#top

Project Description:

This project focuses on the development of a Turkish question-answering assistant based on Retrieval-Augmented Generation (RAG). The system is designed to answer questions using information retrieved from documents, provide references to the sources of its answers, and indicate when sufficient information is not available. The goal is to provide more reliable and transparent answers by allowing users to verify the information against the original documents. By working with Turkish documents, the challenges related to Turkish language characteristics and their effects on answer quality will be investigated. The project aims to contribute to Turkish natural language processing by identifying effective RAG approaches.

Different RAG configurations will be tested and compared using document chunking strategies, Turkish embedding models, vector databases, hybrid search, and re-ranking techniques with open-source language models. System performance will be evaluated based on answer accuracy, source consistency, hallucination rate, and response time. The results will be used to determine which methods work better with Turkish documents. Through this process, a reliable and source-based question-answering assistant will be developed, and the challenges of applying RAG techniques to Turkish contents will be explored.